

2. **Shared KV Fetching.** In the inner loop, Sequentially load continuous key/value blocks indexed by $\mathcal{I}_t$ into SRAM as $K \in \mathbb{R}^{[B_k, d_k]}, V \in \mathbb{R}^{[B_k, d_v]}$ to minimize memory loading, where B_k is the kernel block size satisfying $B_k | l'$.
3. **Outer Loop on Grid.** Since the inner-loop length (proportional to the selected block count n) remains nearly identical for different query blocks, we put query/output loops in Triton’s grid scheduler to simplify and optimize the kernel.

This design achieves near-optimal arithmetic intensity by (1) eliminating redundant KV transfers through group-wise sharing, and (2) balancing compute workloads across GPU streaming multiprocessors.

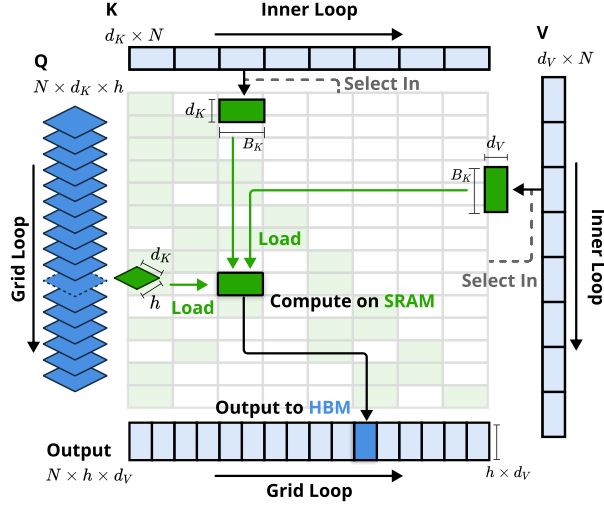


Figure 3 | Kernel design for NSA. The kernel loads queries by GQA groups (Grid Loop), fetches corresponding sparse KV blocks (Inner Loop), and performs attention computation on SRAM. Green blocks indicate data on SRAM, while blue indicates data on HBM.

4. Experiments

We evaluate NSA through three lenses: (1) general benchmarks performance, (2) long-context benchmarks performance, and (3) chain-of-thought reasoning performance, comparing against Full Attention baseline and state-of-the-art sparse attention methods. We defer the efficiency analysis of our sparse computation paradigm to Section 5, where we provide detailed discussions on training and inference speed.

4.1. Pretraining Setup

Following the common practice in state-of-the-art LLMs, our experiments adopt a backbone combining Grouped-Query Attention (GQA) and Mixture-of-Experts (MoE), featuring 27B total parameters with 3B active parameters. The model consists of 30 layers with a hidden dimension of 2560. For GQA, we set the number of groups to 4, with a total of 64 attention heads. For each head, the hidden dimensions of the query, key, and value are configured as $d_q = d_k = 192$ and $d_v = 128$, respectively. For MoE, we utilize the DeepSeekMoE (Dai et al., 2024; DeepSeek-AI, 2024) structure, with 72 routed experts and 2 shared experts, and set the top-k experts to 6. To ensure training stability, the MoE in the first layer is replaced by an MLP in the form of SwiGLU.

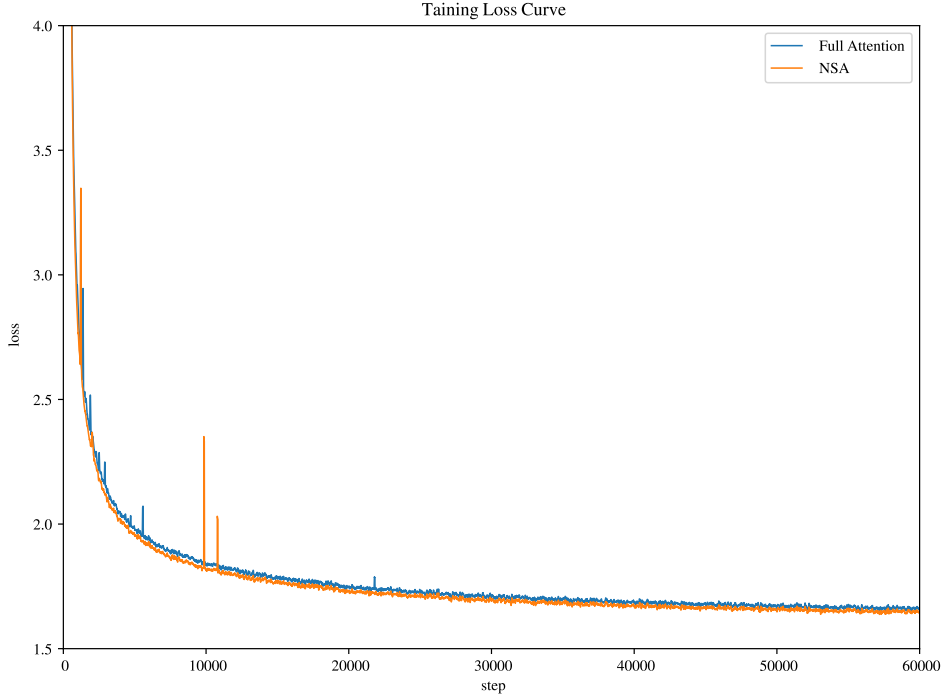


Figure 4 | Pretraining loss comparison between Full Attention and our NSA on 27B-parameter model. Both models exhibit stable convergence, with NSA achieving lower loss values.

Model	MMLU Acc. 5-shot	MMLU-PRO Acc. 5-shot	CMMLU Acc. 5-shot	BBH Acc. 3-shot	GSM8K Acc. 8-shot	MATH Acc. 4-shot	DROP F1 1-shot	MBPP Pass@1 3-shot	HumanEval Pass@1 0-shot	Avg.
Full Attn	0.567	0.279	0.576	0.497	0.486	0.263	0.503	0.482	0.335	0.443
NSA	0.565	0.286	0.587	0.521	0.520	0.264	0.545	0.466	0.348	0.456

Table 1 | Pretraining performance comparison between the full attention baseline and NSA on general benchmarks, across knowledge (MMLU, MMLU-PRO, CMMLU), reasoning (BBH, GSM8K, MATH, DROP), and coding (MBPP, HumanEval) tasks. NSA achieves superior average performance on most benchmarks despite high sparsity.

The proposed architecture achieves an effective trade-off between computation cost and model performance. For NSA, we set compression block size $l = 32$, sliding stride $d = 16$, selected block size $l' = 64$, selected block count $n = 16$ (including fixed activating the 1 initial block and 2 local blocks), and sliding window size $w = 512$. Both Full Attention and sparse attention models are pretrained on 270B tokens of 8k-length texts, followed by continued training and supervised fine-tuning on 32k-length texts with YaRN (Peng et al., 2024) to achieve long-context adaptation. Both models are trained to full convergence to ensure fair comparison. As shown in Figure 4, the pretraining loss curve of our NSA and Full Attention baseline demonstrates stable and smooth decline, with NSA consistently outperforming the Full Attention model.

4.2. Baselines Methods

In addition to comparing with Full Attention, we evaluate several state-of-the-art inference-stage sparse attention methods: H2O (Zhang et al., 2023b), infLLM (Xiao et al., 2024a), Quest (Tang et al., 2024), and Exact-Top, which first computes full attention score and select the top- n scores keys corresponding to each query and then calculates attention on these positions. These

Model	SQA			MQA				Synthetic		Code	Avg.
	MFQA-en	MFQA-zh	Qasper	HPQ	2Wiki	GovRpt	Dur	PassR-en	PassR-zh	LCC	
H2O	0.428	0.429	0.308	0.112	0.101	0.231	0.208	0.704	0.421	0.092	0.303
InfLLM	0.474	0.517	0.356	0.306	0.250	0.277	0.257	0.766	0.486	0.143	0.383
Quest	0.495	0.561	0.365	0.295	0.245	0.293	0.257	0.792	0.478	0.135	0.392
Exact-Top	0.502	0.605	0.397	0.321	0.288	<u>0.316</u>	0.291	0.810	0.548	0.156	0.423
Full Attn	0.512	<u>0.623</u>	<u>0.409</u>	<u>0.350</u>	<u>0.305</u>	0.324	<u>0.294</u>	<u>0.830</u>	0.560	<u>0.163</u>	<u>0.437</u>
NSA	<u>0.503</u>	0.624	0.432	0.437	0.356	0.307	0.341	0.905	<u>0.550</u>	0.232	0.469

Table 2 | Performance comparison between our NSA and baselines on LongBench, including subsets in single document QA, multi-document QA, synthetic and code task categories. NSA outperformed most of the baselines including Full Attention.

methods span diverse sparse attention paradigms, including KV-cache eviction, query-aware selection, and exact top- n sparse selection.

For general evaluation, where most samples have lengths within the local context window of sparse attention baselines, these methods are effectively equivalent to Full Attention. Therefore, we present only the comparison results between NSA and Full Attention baseline in this setting. In the long-context evaluation, we conduct comparisons across all baseline methods, with the sparsity of all sparse attention methods set to the same to ensure a fair comparison. For chain-of-thought reasoning evaluation, which requires long-text supervised fine-tuning, we limit our comparison to Full Attention, as sparse attention baselines do not support training.

4.3. Performance Comparison

General Evaluation. We evaluated the pretrained NSA and Full Attention baseline, on a comprehensive suite of benchmarks spanning knowledge, reasoning, and coding capabilities, including MMLU (Hendrycks et al., 2020), MMLU-PRO (Wang et al., 2024), CMMLU (Li et al., 2023), BBH (Suzgun et al., 2022), GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2020), DROP (Dua et al., 2019), MBPP (Austin et al., 2021), and HumanEval (Chen et al., 2021). The results are shown in Table 1. Despite its sparsity, NSA achieves superior overall performance, outperforming all baselines including Full Attention on 7 out of 9 metrics. This indicates that although NSA may not fully leverage its efficiency advantages on shorter sequences, it shows strong performance. Notably, NSA demonstrates significant gains in reasoning-related benchmarks (DROP: +0.042, GSM8K: +0.034), suggesting that our pretraining helps models to develop specialized attention mechanisms. This sparse attention pretraining mechanism forces model to focus on the most important information, potentially enhancing performance by filtering out noise from irrelevant attention pathways. The consistent performance across diverse evaluations also validates NSA’s robustness as a general-purpose architecture.

Long-Context Evaluation. As shown in Figure 5, NSA achieves perfect retrieval accuracy across all positions in 64k-context needle-in-a-haystack (Kamradt, 2023) test. This performance stems from our hierarchical sparse attention design, which combines compression tokens for efficient global context scanning, and selection tokens for precise local information retrieval. The coarse-grained compression identifies relevant context blocks at low computational cost, while the token-level attention on selected tokens ensures the preservation of critical fine-grained information. This design enables NSA to maintain both global awareness and local precision.

We also evaluate NSA on LongBench (Bai et al., 2023) against state-of-the-art sparse attention

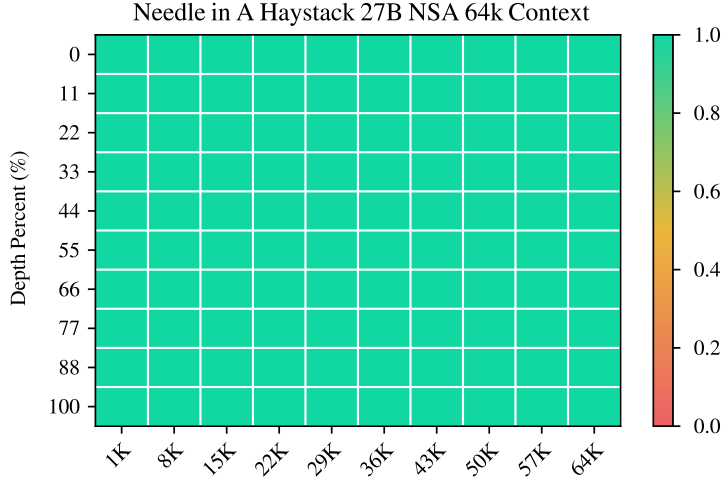


Figure 5 | Needle-in-a-Haystack retrieval accuracy across context positions with 64k context length. NSA achieves perfect accuracy through its hierarchical sparse attention design.

methods and Full Attention baseline. To ensure consistent sparsity, we set the token activated by each query in all sparse attention baselines to 2560 tokens, which corresponds to the average number of tokens activated in NSA when handling 32k sequence lengths. Following StreamLLM (Xiao et al., 2023), this token budget includes the leading 128 tokens and 512 local tokens. We exclude certain subsets from LongBench due to their low scores across all models, which may not provide meaningful comparisons. As shown in Table 2, NSA achieves the highest average score 0.469, outperforming all baselines (+0.032 over Full Attention and +0.046 over Exact-Top). This improvement arises from two key innovations: (1) our native sparse attention design, which enables end-to-end optimization of sparse patterns during pretraining, facilitates synchronized adaptation between the sparse attention module and other model components; and (2) the hierarchical sparse attention mechanism achieves a balance between local and global information processing.

Notably, NSA demonstrates exceptional performance on tasks requiring complex reasoning over long contexts, achieving +0.087 and +0.051 improvements over Full Attention on multi-hop QA tasks (HPQ and 2Wiki), exceeding the performance of baselines on code understanding (LCC: +0.069), and outperforming other methods on passage retrieval (PassR-en: +0.075). These results validate NSA’s capability to handle diverse long-context challenges, with its natively pretrained sparse attention providing additional benefits in learning task-optimal patterns.

Chain-of-Thought Reasoning Evaluation. To evaluate NSA’s compatibility with advanced downstream training paradigms, we investigate its capacity to acquire chain-of-thought mathematical reasoning abilities via post-training. Given the limited effectiveness of reinforcement learning on smaller-scale models, we employ knowledge distillation from DeepSeek-R1, conducting supervised fine-tuning (SFT) with 10B tokens of 32k-length mathematical reasoning traces. This produces two comparable models: Full Attention-R (Full Attention baseline) and NSA-R (our sparse variant). We assess both models on the challenging American Invitational Mathematics Examination (AIME 24) benchmark. We use a sampling temperature of 0.7 and a top- p value of 0.95 to generate 16 responses for each question and obtain the average score. To validate the impact of reasoning depth, we conduct experiments with two generation context limits: 8k and 16k tokens, measuring whether extended reasoning chains improve accuracy. Example comparisons of model predictions are provided in Appendix A.